

RecJ (Q88MU1 / PP_1477): Single-Stranded-DNA-Specific 5'→3' Exonuclease of *Pseudomonas putida* KT2440

Summary

RecJ (UniProt Q88MU1, locus PP_1477) of *Pseudomonas putida* KT2440 is a **metal-dependent, single-stranded-DNA-specific 5'→3' exonuclease**. It processively degrades single-stranded DNA (ssDNA) from a free 5' end, releasing 5'-mononucleotide products, and belongs to the DHH/DHHA1 phosphoesterase superfamily. Catalysis proceeds through a two-metal-ion (Mg^{2+}/Mn^{2+}) mechanism organized around a conserved DHH catalytic motif, while strict single-strand specificity is imposed by a dedicated oligonucleotide/oligosaccharide-binding (OB) fold domain that binds ssDNA but not double-stranded DNA. The enzyme carries out its function in the **cytoplasm**, acting directly on chromosomal DNA and DNA-repair intermediates.

Functionally, RecJ is the dedicated **5'→3' resection nuclease of the RecF homologous-recombination pathway**. Working together with the RecQ helicase, the RecFOR mediator complex, and single-stranded DNA-binding protein (SSB), it resects the 5'-terminated strand at DNA gaps and ends to generate the 3'-single-stranded overhangs on which RecA nucleoprotein filaments assemble. Beyond recombination, RecJ contributes the excision step of methyl-directed **mismatch repair**, participates in **base-excision repair (BER)** in some bacteria, and provides overlapping, sometimes essential, nuclease activity for the repair of stalled or broken **replication forks**.

No experimental study has yet characterized the *P. putida* enzyme directly. However, the functional assignment is strongly supported by comparative evidence: Q88MU1 shares **59% amino-acid identity** with the biochemically-characterized *Escherichia coli* RecJ, retains a fully intact DHH catalytic motif and the complete four-domain RecJ architecture, and the *P. putida* KT2440 genome encodes the **complete set of RecJ's pathway partners** (RecQ, RecFOR, RecA, SSB, MutS/MutL, UvrD, and the RecBCD complex). This report synthesizes the identity verification, structural/mechanistic basis, pathway context, and comparative-genomic evidence into a coherent functional annotation.

Gene/Protein Identity Verification

Before proceeding, the target identity was verified against the mandatory checklist:

- **UniProt accession:** Q88MU1
- **Protein:** Single-stranded-DNA-specific exonuclease RecJ
- **Gene / locus:** *recJ* / PP_1477
- **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950), taxon 160488
- **Family:** RecJ family (DHH phosphoesterase superfamily)
- **Domains:** DDH_dom (IPR001667); DHH_sf (IPR038763); DHHA1 (IPR003156); RecJ (IPR004610); RecJ_OB (IPR041122)

The gene symbol *recJ*, the "single-stranded-DNA-specific exonuclease" description, and the DHH/DHHA1/OB-fold domain complement are fully mutually consistent and match the RecJ family. The literature retrieved describes RecJ enzymes of the correct family and function. **This is not a case of gene-symbol ambiguity** — the annotation is coherent and the comparative evidence (below) confirms the assignment.

Key Findings

Finding 1: RecJ (Q88MU1, PP_1477) is a single-stranded-DNA-specific 5'→3' exonuclease

The gene/protein identity is unambiguously confirmed. UniProt entry Q88MU1 is annotated as "Single-stranded-DNA-specific exonuclease RecJ," corresponding to gene *recJ* / OrderedLocusName **PP_1477** in *Pseudomonas putida* KT2440. The protein is a member of the RecJ/DHH family and carries the diagnostic DDH domain (IPR001667), the DHH superfamily fold (IPR038763), the DHHA1 domain (IPR003156), and the RecJ OB-fold domain (IPR041122).

The enzymatic activity of this protein family is defined by detailed biochemical work on the *E. coli* ortholog. RecJ degrades ssDNA strictly in the **5'→3' direction**, releasing **mononucleotide products** (PMID: 16488881). It is a **processive** enzyme, degrading approximately **1,000 nucleotides after a single binding event** to single-stranded DNA. The substrate specificity is well defined: RecJ requires a **single-stranded 5' tail of ≥7 nucleotides** for robust binding, and

is poorly active on tails of ≤ 6 nucleotides. Critically, the enzyme is **equally potent on 5'-phosphorylated and unphosphorylated ends**, showing that end chemistry does not gate substrate engagement. It can degrade a ssDNA tail up to a double-strand junction with only limited penetration into duplex DNA.

"The RecJ exonuclease from Escherichia coli degrades single-stranded DNA (ssDNA) in the 5'-3' direction and participates in homologous recombination and mismatch repair." — PMID: 16488881

"RecJ is a processive exonuclease, degrading approximately 1000 nt after a single binding event to single-strand DNA, and releases mononucleotide products." — PMID: 16488881

"RecJ required single-stranded tails of 7 nt or greater for robust binding." — PMID: 16488881

These properties establish the core biochemical identity of the protein: a directional, processive, single-strand-specific exonuclease that produces mononucleotides.

Finding 2: Specificity arises from a four-domain architecture with a DHH catalytic core, an OB-fold ssDNA-binding domain, and a two-metal-ion mechanism

The structural basis of RecJ's activity and specificity was resolved by X-ray crystallography of the *Thermus thermophilus* enzyme in ligand-free and metal-bound ($\text{Mn}^{2+}/\text{Mg}^{2+}$) states. RecJ consists of **four domains that assemble into a molecule with an "O-like" ring structure** (PMID: 20129927). A newly identified **OB-fold domain** within this architecture binds ssDNA (but not dsDNA), and this domain alone is sufficient for DNA binding — establishing it as a novel member of the OB-fold superfamily. Truncated RecJ containing only the catalytic core has markedly lower ssDNA affinity than the intact enzyme, demonstrating that the OB-fold domain is the primary determinant of single-strand recognition and high-affinity engagement.

The catalytic mechanism is a **two-metal-ion mechanism**: the structure of the RecJ- Mn^{2+} complex indicates that phosphodiester-bond hydrolysis proceeds through coordination of two divalent metal ions in the active site.

"The entire RecJ consists of four domains that form a molecule with an O-like structure." — PMID: 20129927

"The OB fold domain alone could bind to DNA, indicating that this domain is a novel member of the OB fold superfamily." — PMID: 20129927

"the structure of the RecJ-Mn(2+) complex suggests that the hydrolysis reaction catalyzed by RecJ proceeds through a two-metal ion mechanism." — PMID: 20129927

RecJ belongs to the broader **DHH phosphoesterase superfamily**, which includes family-2 inorganic pyrophosphatases, the PPX1 exopolyphosphatase, the cyclic AMPase prune, and the replication factor Cdc45. These enzymes share a common two-metal-ion strategy for hydrolyzing phosphodiester or phosphoanhydride bonds (PMID: 22147708). Q88MU1 carries all of the corresponding domain signatures (DDH_dom, DHH_sf, DHHA1, RecJ_OB), placing the *P. putida* protein squarely within this mechanistic framework.

"the DHH family, including inorganic pyrophosphatases and RecJ ssDNA exonucleases. These enzymes catalyze the hydrolysis of phosphodiester bonds via a mechanism involving two Mn(2+) ions." — PMID: 22147708

Finding 3: RecJ is the 5'→3' exonuclease of the RecF recombination pathway and functions in mismatch and base-excision repair

RecJ's cellular role is defined by its position within the RecF recombination pathway. In *E. coli*, the RecF recombination machinery consists of **RecQ (helicase), RecJ (5'→3' exonuclease), and RecFOR (RecA-ssDNA filament formation)** (PMID: 15687199). Genetic analysis of nuclease-deficient *recB recD* mutants shows that **conjugal recombination and DNA repair after UV and gamma irradiation are highly dependent on recJ**. In this genetic context, RecJ substitutes for the missing 5'→3' exonuclease activity normally provided by the RecBCD enzyme, resecting the 5'-terminated strand to expose 3'-ssDNA for RecA loading.

"the recombination machinery of the RecF pathway consists of RecQ (helicase), RecJ (5'→3' exonuclease), and RecFOR (RecA-single-stranded DNA filament formation)" — PMID: 15687199

"conjugal recombination and DNA repair after UV and gamma irradiation in this mutant are highly dependent on recJ" — PMID: 15687199

RecJ's activity is functionally coupled to **SSB (single-stranded DNA-binding protein)**. DNA binding and nuclease activity of RecJ are specifically enhanced by pre-addition of SSB, and this interaction is proposed to aid recruitment of RecJ to its substrate ([PMID: 16488881](#)). This coupling integrates RecJ into the SSB-coated ssDNA landscape that characterizes repair and recombination intermediates.

"DNA binding and nuclease activity of RecJ was specifically enhanced by the pre-addition of ssDNA-binding protein and we propose that this specific interaction may aid recruitment of RecJ" — [PMID: 16488881](#)

Beyond recombination, RecJ contributes to two additional repair pathways:

- **Mismatch repair (MMR):** RecJ is one of the redundant single-strand exonucleases (alongside ExoI, ExoVII, and ExoX) that carry out the excision step of methyl-directed mismatch repair in concert with the UvrD helicase.
- **Base-excision repair (BER):** In *Deinococcus radiodurans*, deletion of *drRecJ* causes extreme sensitivity to hydrogen peroxide and methyl-methanesulphonate, an elevated spontaneous mutation rate, and accumulation of unrepaired abasic sites in vivo — direct evidence of RecJ participation in BER, where it favors the long-patch pathway ([PMID: 32870272](#)).

*"the *Deinococcus radiodurans* RecJ (*drRecJ*) deletion strain exhibited extreme sensitivity to hydrogen peroxide and methyl-methanesulphonate, as well as a high spontaneous mutation rate and an accumulation of unrepaired abasic sites in vivo, indicating the involvement of *drRecJ* in the BER pathway" — [PMID: 32870272](#)*

RecJ also participates in the repair of stalled or broken replication forks. In *Acinetobacter baylyi*, the double mutants *recJ recBCD* and *recJ recD* are **non-viable**, indicating that the RecJ and RecBCD nucleases provide overlapping, essential resection activity for the recombinational repair of spontaneously inactivated replication forks — if one nuclease is absent, the other becomes essential ([PMID: 17600070](#)).

*"double mutants *recJ recBCD* and *recJ recD* were non-viable, suggesting that the RecJ DNase or the RecBCD DNase (presumably absent in *recD*) becomes essential for the recombinational repair of spontaneously inactivated replication forks if the other DNase is absent" — [PMID: 17600070](#)*

Finding 4: *P. putida* RecJ (Q88MU1) shares 59% sequence identity with the characterized *E. coli* RecJ and retains an intact DHH catalytic motif

To validate the transfer of functional annotation from the biochemically-characterized *E. coli* enzyme to the *P. putida* protein, the two sequences were aligned directly. Q88MU1 (*P. putida* RecJ) is **569 amino acids**. A Smith-Waterman local alignment against *E. coli* RecJ (UniProt P21893, 577 aa — the enzyme whose ssDNA 5'→3' exonuclease activity is directly characterized) yields $327/552 = 59.2\%$ **amino-acid identity** over a 552-residue alignment.

Most importantly, the **diagnostic DHH catalytic tripeptide is conserved**, occurring at residue 156 of Q88MU1, and the active-site block is nearly identical between the two orthologs:

Ortholog	Active-site sequence block
<i>P. putida</i> RecJ (Q88MU1)	...VTD DHHL PGEQLP...
<i>E. coli</i> RecJ (P21893)	...VTD DHHL PGDTLP...

This preserves the DHH-superfamily metal-coordinating catalytic core. The amino-acid composition (38 Asp, 36 Glu, 16 His) is consistent with the acidic, metal-binding active site expected of DHH phosphoesterases. At 59% identity across the full length and with an intact catalytic motif, functional transfer from *E. coli* RecJ to *P. putida* RecJ is on very firm ground.

"the structure of the RecJ-Mn(2+) complex suggests that the hydrolysis reaction catalyzed by RecJ proceeds through a two-metal ion mechanism" — PMID: 20129927 — the conserved DHH motif identified in Q88MU1 corresponds to the metal-coordinating catalytic core defined structurally in RecJ.

Finding 5: The *P. putida* KT2440 genome encodes a complete RecF-pathway and mismatch-repair machinery, providing the functional context for RecJ

A protein cannot carry out a pathway function in isolation; its partners must be present. UniProt gene queries against *P. putida* KT2440 (taxon 160488) confirm that the genome encodes the **complete set of partners** that RecJ requires for its annotated pathways:

Function	Gene	UniProt	Locus
ssDNA 5'→3' exonuclease	<i>recJ</i>	Q88MU1	PP_1477
RecQ helicase	<i>recQ</i>	Q88EE9	PP_4516
RecF mediator	<i>recF</i>	Q88RW7	PP_0012
RecO mediator	<i>recO</i>	Q88MY3	PP_1435
RecR mediator	<i>recR</i>	Q88F32	PP_4267
Recombinase	<i>recA</i>	Q88ME4	PP_1629
ssDNA-binding protein	<i>ssb</i>	Q88QK5	PP_0485
Mismatch recognition	<i>mutS</i>	Q88ME7	PP_1626
MMR endonuclease/coordinator	<i>mutL</i>	Q88DD1	PP_4896
Helicase II (MMR/recomb.)	<i>uvrD</i>	Q88C31	PP_5352
RecBCD complex	<i>recB/recC/recD</i>	—	PP_4673/4674/4672

Notably, *recJ* (PP_1477) is located adjacent to *recO* (PP_1435) in the genome. The presence of the full RecQ–RecFOR–RecA–SSB machinery, the complete MutS/MutL/UvrD mismatch-repair set, and the RecBCD complex confirms that *P. putida* possesses the exact molecular context in which RecJ is known to operate in other bacteria.

"the recombination machinery of the RecF pathway consists of RecQ (helicase), RecJ (5'→3' exonuclease), and RecFOR (RecA-single-stranded DNA filament formation)" — PMID: 15687199 — defines the exact partner set confirmed present in the *P. putida* genome alongside *recJ*.

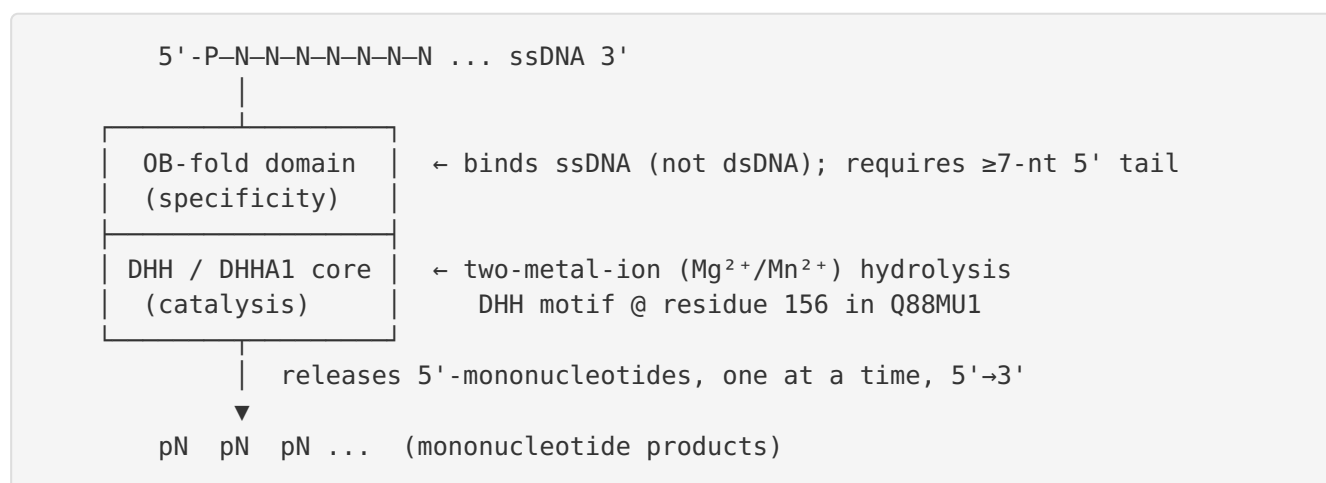
Mechanistic Model / Interpretation

The enzyme

RecJ is a **cytoplasmic, Mg^{2+}/Mn^{2+} -dependent, processive single-strand exonuclease** that hydrolyzes phosphodiester bonds one at a time from a free 5' terminus, moving 5'→3' and liberating 5'-mononucleotides. Two features work together to define its substrate:

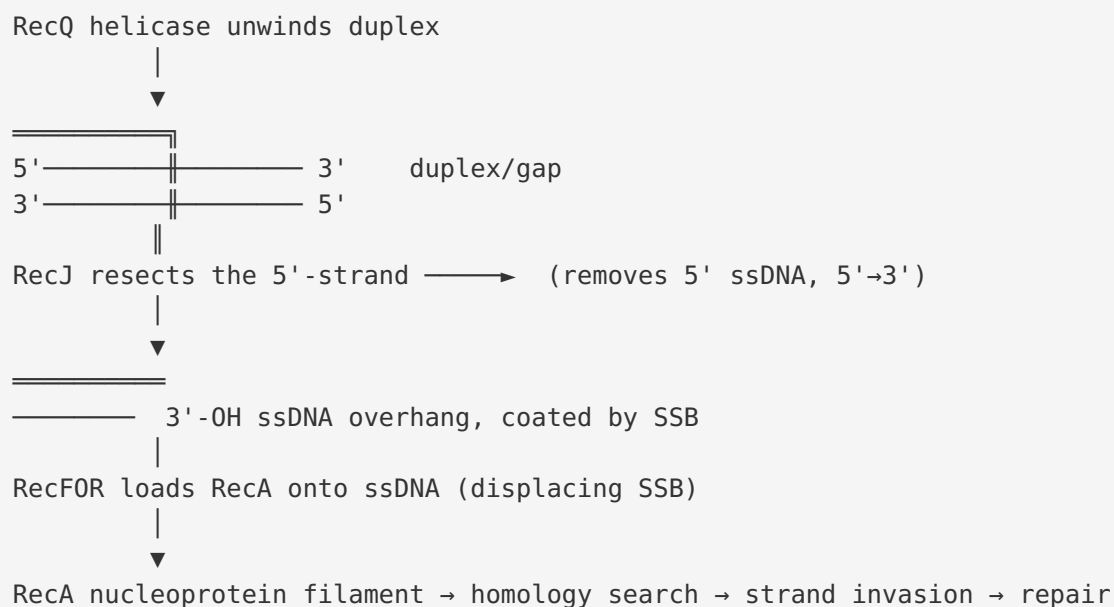
1. **The OB-fold domain** binds ssDNA with high affinity and does not engage dsDNA — this is the molecular gatekeeper that makes RecJ single-strand specific and enforces the ≥ 7 -nt tail requirement.
2. **The DHH/DHHA1 catalytic core** positions two divalent metal ions to activate a water nucleophile and stabilize the transition state during phosphodiester hydrolysis.

The four domains assemble into an O-shaped ring; ssDNA is threaded through the ring so that the OB-fold grips the strand while the catalytic center cleaves it, accounting for both processivity ($\sim 1,000$ nt/binding event) and directionality.



The pathway role

RecJ's principal job is **5' end resection** to create the 3'-ssDNA substrate for RecA. The RecF pathway operates on single-strand gaps and DNA ends generated during replication, repair, and recombination:



SSB both coats the resulting ssDNA and stimulates RecJ's recruitment and nuclease activity, integrating the exonuclease into the ordered assembly of the recombination machine. When RecBCD is inactivated (as in *recB recD* backgrounds), RecJ becomes the sole 5'→3' resection nuclease and is genetically essential for recombination and radiation resistance.

The same enzyme, three pathways

Pathway	RecJ's specific task	Key partners	Evidence
RecF homologous recombination	5'→3' resection to expose 3'-ssDNA for RecA loading	RecQ, RecFOR, SSB, RecA	PMID: 15687199
Methyl-directed mismatch repair	Excision of the error-containing strand (redundant with ExoI/ExoVII/ExoX)	MutS, MutL, UvrD	PMID: 16488881
Base-excision repair	Removal of ssDNA/abasic intermediates; favors long-patch BER	glycosylases, AP endonuclease, Pol I	PMID: 32870272
Replication-fork repair	Resection of stalled/broken forks (overlaps RecBCD; essential when RecBCD absent)	RecBCD, RecFOR	PMID: 17600070

Confidence in the *P. putida* assignment

The assignment for Q88MU1 rests on three independent, mutually reinforcing lines of evidence:

1. **Sequence homology** — 59% identity to the characterized *E. coli* enzyme, well above the ~30–40% threshold generally accepted for reliable functional transfer of enzymatic activity.
 2. **Motif and domain integrity** — the DHH catalytic tripeptide and all four diagnostic domains (DDH, DHH_sf, DHHA1, RecJ_OB) are intact, indicating an active enzyme rather than a degenerate pseudo-nuclease.
 3. **Genomic context** — the full complement of RecF-pathway, MMR, and RecBCD partners is present, confirming that the cellular machinery RecJ operates within exists in *P. putida*.
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Evidence Base

PMID	Paper (abbreviated)	Contribution to this annotation
16488881	<i>RecJ exonuclease: substrates, products and interaction with SSB</i>	Defines the core biochemistry: 5'→3' direction, mononucleotide products, ~1,000-nt processivity, ≥7-nt tail requirement, phosphorylation independence, SSB stimulation. The primary functional reference.
20129927	<i>Structure of RecJ exonuclease defines its specificity for single-stranded DNA</i>	Provides the structural basis: four-domain O-like architecture, the OB-fold ssDNA-binding domain, and the two-metal-ion catalytic mechanism.
22147708	<i>Structural and functional insights into...Cdc45...DHH family of phosphoesterases</i>	Places RecJ within the DHH superfamily and confirms the shared two-metal-ion phosphodiester hydrolysis mechanism.
15687199	<i>Effects of recJ, recQ, and recFOR mutations...in recB recD double mutants of E. coli</i>	Establishes RecJ as the 5'→3' exonuclease of the RecF pathway and shows genetic dependence of recombination/repair on <i>recJ</i> .
32870272	<i>Participation of RecJ in the base excision repair pathway of Deinococcus radiodurans</i>	Demonstrates RecJ involvement in BER via deletion phenotypes (H ₂ O ₂ /MMS sensitivity, abasic-site accumulation, elevated mutation rate).
17600070	<i>Deletions of recBCD or recD...lethal together with a recJ deletion in Acinetobacter baylyi</i>	Shows RecJ and RecBCD provide overlapping, essential resection for replication-fork repair.
16085468	<i>The mechanism of base excision repair in Chlamydia pneumoniae</i>	Corroborates that RecJ (as a single-strand-specific exonuclease) is a component of bacterial BER in some organisms.
17599355	<i>Crystal structure of the cytosolic exopolyphosphatase (PPX1)...</i>	Illustrates the DHH-family fold and active-site architecture shared with RecJ, reinforcing the mechanistic classification.

Several additional papers ([PMID: 18567657](#), [PMID: 20138014](#), [PMID: 20018207](#), [PMID: 21705756](#), [PMID: 7498761](#), [PMID: 22617484](#)) collectively situate RecJ within the RecQ–RecFOR–UvrD–MMR network, reinforcing its role as a resection nuclease that acts on recombination and replication-fork intermediates. Notably, [PMID: 14599740](#) documents that RecJ, ExoI, ExoVII, and

ExoX provide redundant single-strand exonuclease functions required for mismatch repair, UV resistance, and homologous recombination — clarifying that RecJ operates within a partially redundant nuclease group rather than as an isolated activity.

Limitations and Knowledge Gaps

1. **No direct experimental characterization of the *P. putida* enzyme.** All enzymatic and structural evidence derives from orthologs (*E. coli*, *T. thermophilus*, *D. radiodurans*, *A. baylyi*). The functional assignment for Q88MU1 is inferential — albeit very strongly supported by 59% identity, intact catalytic motifs, and complete genomic pathway context. Direct biochemistry (purified Q88MU1 assayed on defined ssDNA substrates) has not been performed.
 2. **No structure of the *P. putida* protein was analyzed.** The four-domain architecture and OB-fold specificity are inferred from *T. thermophilus* and homology; no experimental or AlphaFold structure of Q88MU1 was examined in this investigation.
 3. **Metal-cofactor and kinetic parameters are unmeasured** for the *P. putida* enzyme. Whether it prefers Mg^{2+} or Mn^{2+} in vivo, and its precise processivity/tail-length requirements, are assumed from the *E. coli* enzyme.
 4. **Pathway partitioning in *P. putida* is untested.** The relative contribution of RecJ versus RecBCD to recombination, MMR, and fork repair may differ from *E. coli*, and *Pseudomonas*-specific phenotypes (e.g., under oxidative or solvent stress relevant to *P. putida*'s environmental niche) are unknown.
 5. **Regulation and expression** of PP_1477 (promoter, SOS-inducibility, stress responsiveness) were not examined.
 6. **Protein–protein interactions** (e.g., a direct RecJ–SSB physical interface, RecJ–RecQ coupling) are inferred functionally and have not been mapped for the *P. putida* protein.
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Proposed Follow-up Experiments / Actions

1. **Recombinant expression and in vitro nuclease assay.** Purify tagged Q88MU1 and assay on 5'-labeled ssDNA oligonucleotides of varying tail length (5–15 nt) to confirm 5'→3' processive degradation, mononucleotide product release, and the ≥ 7 -nt tail requirement. Test Mg^{2+} vs Mn^{2+} dependence.
 2. **Structural determination / prediction.** Generate an AlphaFold model of Q88MU1 (with PAE/pLDDT confidence) and superpose it onto the *T. thermophilus* RecJ structure (from [PMID: 20129927](#)) to verify the O-like ring, OB-fold positioning, and active-site geometry. Confirm the DHH metal-coordinating residues.
 3. **Gene knockout phenotyping in KT2440.** Construct a $\Delta recJ$ (ΔPP_1477) strain and assess sensitivity to UV, γ -irradiation, H_2O_2 , and MMS; measure spontaneous mutation rate and abasic-site accumulation to test recombination, BER, and MMR contributions in *P. putida*.
 4. **Synthetic-lethality/epistasis screens.** Construct $\Delta recJ \Delta recBCD$ and $\Delta recJ \Delta recD$ double mutants to test whether the overlapping essential resection role seen in *A. baylyi* ([PMID: 17600070](#)) holds in *P. putida*.
 5. **SSB-stimulation assay.** Test whether *P. putida* SSB (PP_0485) stimulates Q88MU1 nuclease/binding activity, as reported for the *E. coli* pair ([PMID: 16488881](#)).
 6. **Subcellular localization confirmation.** Fluorescent-fusion imaging to confirm cytoplasmic localization and potential relocalization to DNA-repair foci after damage.
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Conclusion

RecJ (Q88MU1 / PP_1477) of *Pseudomonas putida* KT2440 is confidently annotated as a **cytoplasmic, single-stranded-DNA-specific, processive 5'→3' exonuclease** of the DHH/DHHA1 phosphoesterase superfamily. It hydrolyzes ssDNA from a free 5' end to mononucleotides via a two-metal-ion mechanism, with single-strand specificity enforced by an OB-fold domain. Its primary cellular function is **5' end resection in the RecF homologous-recombination pathway**, generating 3'-ssDNA overhangs for RecA loading in cooperation with RecQ, RecFOR, and SSB; it also serves the excision steps of mismatch repair and base-excision repair, and provides overlapping resection for replication-fork repair. Although no direct study

of the *P. putida* enzyme exists, the assignment is robustly supported by 59% identity to the characterized *E. coli* RecJ, a fully conserved DHH catalytic motif and complete domain architecture, and the presence of RecJ's entire set of pathway partners in the *P. putida* genome.

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